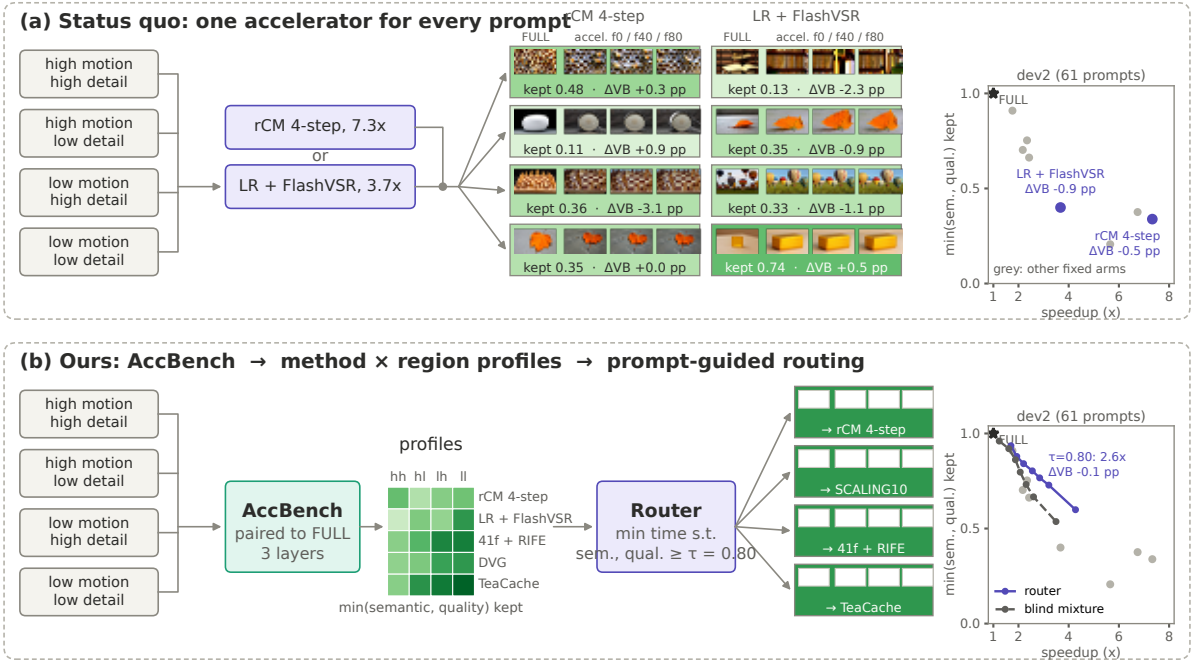

(a) Status quo: one accelerator for every prompt
high motion high detail
high motion low detail
low motion high detail
low motion low detail
rCM 4-step, 7.3x
or
LR + FlashVSR, 3.7x
CM 4-step
LR + FlashVSR
FULL
accel. f0 / f40 / f80
kept 0.48 · ΔVB +0.3 pp
kept 0.13 · ΔVB -2.3 pp
kept 0.11 · ΔVB +0.9 pp
kept 0.35 · ΔVB -0.9 pp
kept 0.36 · ΔVB -3.1 pp
kept 0.33 · ΔVB -1.1 pp
kept 0.35 · ΔVB +0.0 pp
kept 0.74 · ΔVB +0.5 pp
dev2 (61 prompts)
FULL
LR + FlashVSR
ΔVB -0.9 pp
rCM 4-step
ΔVB -0.5 pp
grey: other fixed arms
min(sem., qual.) kept
speedup (x)
(b) Ours: AccBench → method × region profiles → prompt-guided routing
high motion high detail
high motion low detail
low motion high detail
low motion low detail
AccBench
paired to FULL
3 layers
profiles
hh hl lh ll
rCM 4-step
LR + FlashVSR
41f + RIFE
DVG
TeaCache
min(semantic, quality) kept
Router
min time s.t.
sem., qual. ≥ τ = 0.80
→ rCM 4-step
→ SCALING10
→ 41f + RIFE
→ TeaCache
dev2 (61 prompts)
FULL
τ=0.80: 2.6x
ΔVB -0.1 pp
router
blind mixture
min(sem., qual.) kept
speedup (x)